

The Unaccusative-Unergative Distinction in Mandarin Resultative Constructions

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1. Introduction

The *Unaccusativity Hypothesis* (UH), which posits that the sole argument of an unaccusative verb is a Theme/underlying object, while the sole argument of an unergative verb is an Agent/underlying subject (Perlmutter 1978; Burzio 1986), is supported by unaccusative-unergative distinctions in English (intransitive and transitive) resultatives (see e.g., Levin & Rappaport Hovav 1995; a.o.), but is challenged by the apparent lack of such distinctions in Mandarin (intransitive and transitive) resultatives (see e.g., Huang 2006; a.o.).

In this paper, I show that, despite appearances to the contrary, Mandarin transitive resultatives do exhibit unaccusative–unergative distinctions, with respect to (i) the possibility of having a corresponding BA-construction, which diagnoses affectedness/the structural position of the post-verbal object (following Chen 2023; cf. Huang 1992), and (ii) the possibility of passivization with BEI, which diagnoses agentivity/the presence of an underlying matrix subject (following Chen 2023, 2025). I propose a non-uniform analysis of Mandarin resultatives with unaccusative and unergative matrix verbs, in accordance with UH: In intransitive resultatives, the sole argument is a Causee object when the matrix verb is unaccusative and a Causer subject when the matrix verb is unergative. For transitive resultatives, the canonical argument structure is derivable from the intransitive resultative argument structures, either with an additional Causer subject when the matrix verb is unaccusative or with an additional Causee object when the matrix verb is unergative. In addition, Mandarin allows two non-canonical argument structures, which obscure the canonical unaccusative–unergative distinction.

The proposal has two implications: (i) In Mandarin, despite the possibility of non-canonical argument structures, the sole argument of an unaccusative matrix verb is a Causee object in intransitive resultatives and always aligns with the structurally lower argument in transitive resultatives, while the sole argument of an unergative matrix verb is a Causer subject in intransitive resultatives and always aligns with the structurally higher argument in transitive resultatives. (ii) The Mandarin–English differences are derived from cross-linguistic variation in how the principle of control, case assignment, and the structural position of the Causer are encoded in resultative constructions.

2. Apparent (lack of) unaccusative-unergative distinctions in resultatives

Resultative constructions are characterized by the presence of an embedded resultative predicate. A transitive resultative construction, as schematized in (1a), can be analyzed as a causative construction characterized by (i) bi-eventive semantics, with the matrix verb specifying a causing event and the embedded predicate specifying a caused/resulting event (see, e.g., Dowty 1979; Parsons 1990; Levin & Rappaport Hovav 1995; a.o.), and a bi-clausal syntax, in which the matrix verb modifies a causative head that projects a predicate of the causing event, and the embedded predicate projects a predicate of the caused/resulting event, which is selected by the causative head; (ii) the presence of two additional event participants/arguments, namely, the matrix subject and the post-verbal object; and (iii) subjection to the

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so-called *Direct Object Restriction (DOR)*: the embedded result must predicate over the post-verbal object, and, crucially, not the matrix subject (see e.g., Simpson 1983; Levin & Rappaport Hovav 1995). By contrast, an intransitive resultative construction, as schematized in (1b), lacks a post-verbal object, and the embedded result predicates over the matrix subject.

- (1) a. *Trans. resultative*
 Subj V_{Cause} Obj XP_{Result}
 b. *Intrans. resultative*
 Subj V XP_{Result}

In English, resultatives exhibit apparent unaccusative–unergative distinctions in accordance with *UH* (see e.g., Levin & Rappaport Hovav 1995; a.o.): (i) An intransitive resultative is compatible with an unaccusative but not unergative matrix verb, see e.g., (2a) vs. (2b). (ii) A resultative with an unaccusative matrix verb may transitivize with an additional subject but not object, see e.g., (4a) vs. (3a). Conversely, (iii) a resultative with an unergative matrix verb may transitivize with an additional object but not subject, see e.g., (3b) vs. (4b). These contrasts follow from the argument-structural difference between resultatives with unaccusative and unergative matrix verbs (see e.g., Levin & Rappaport Hovav 1995; a.o.).

- (2) *Intrans. resultative*
 a. ... w/ *unaccus. matrix verb*
 The river [froze] solid.
 b. ... w/ *unerg. matrix verb*
 *Dora [shouted] hoarse.
 (3) *Trans. resultative w/ addt'l post-verbal obj.*
 a. ... w/ *unaccus. matrix verb*
 *The snow [melted] the road slushy.
 b. ... w/ *unerg. matrix verb*
 She [cried] me awake.
 (4) *Trans. resultative w/ addt'l matrix subj.*
 a. ... w/ *unaccus. matrix verb*
 The cold weather [froze] the river solid.
 b. ... w/ *unerg. matrix verb*
 *The bad news [cried] me awake.

Unlike English resultatives, Mandarin resultatives lack apparent unaccusative-unergative distinctions (see e.g., Huang 2006; a.o.):¹ (i) An intransitive resultative is compatible with an unaccusative or unergative matrix verb, see e.g., (5a) and (5b). (ii) A resultative with an unaccusative matrix verb may transitivize with an additional subject or object, see e.g., (7a) and (6a). Similarly, (iii) A resultative with an unergative matrix verb may transitivize with an additional object or subject, see e.g., (6b) and (7b).

- (5) *Intrans. resultative*
 a. ... w/ *unaccus. matrix verb*
 Lisi [ji]-de bing-le.
 Lisi be.worried-DE be.sick-PRF
 'Lisi was worried, as a result (he) was sick.'
 b. ... w/ *unerg. matrix verb*
 Lisi [ku]-de bing-le.
 Lisi cry-DE be.sick-PRF
 'Lisi cried, as a result (he) was sick.'
 (6) *Trans. resultative w/ addt'l post-verbal obj.*
 a. ... w/ *unaccus. matrix verb*
 Wo [jidong]-de lian hong-le.
 1SG be.excited-DE face be.red-PRF
 'I was excited, as a result my face was red.'
 b. ... w/ *unerg. matrix verb*
 Wo [ku]-de lian hong-le.
 1SG cry-DE face be.red-PRF
 'I cried, as a result my face was red.'

¹ In Mandarin, an unaccusative-unergative distinction is expected between unaccusative *psych-predicates* e.g., (*zhao*)*ji* 'be worried', *jidong* 'be excited' and unergative verbs e.g., *ku* 'cry', *xiao* 'laugh'.

- (i) a. *Unaccus. psych-predicate construction*
 Lisi hen [zhaoji/jidong].
 Lisi DEG be.worried/be.excited
 'Lisi was worried/excited.'
 b. *Unerg. construction*
 Lisi [ku/xiao]-le.
 Lisi cry/laugh-PRF
 'Lisi cried/laughed.'

(7) *Trans. resultative w/ add'l matrix subj.*

- a. ... w/ *unaccus. matrix verb*
Zhe-shi [ji]-de Lisi bing-le.
 this-matter be.worried-DE Lisi be.sick-PRF
 'This matter made Lisi worry, as a result (he) was sick.'
- b. ... w/ *unerg. matrix verb*
Zhe-shi [ku]-de Lisi bing-le.
 this-matter cry-DE Lisi be.sick-PRF
 'This matter made Lisi cry, as a result (he) was sick.'

To summarize, the unaccusative–unergative distinctions in English transitive resultatives and the apparent lack of such distinctions in Mandarin transitive resultatives are manifested in the mapping between the sole argument of the unaccusative or unergative matrix verb and different arguments in the transitive resultative: In English, this mapping is one-to-one, whereas in Mandarin, this mapping is flexible. Specifically, both languages exhibit a *canonical* mapping, in which the Theme of an unaccusative matrix verb aligns with the post-verbal object and the Agent of an unergative matrix verb aligns with the matrix subject. However, Mandarin but not English also allows a reversed, *non-canonical* mapping, in which the Theme of an unaccusative matrix verb aligns with the post-verbal object and the Agent of an unergative matrix verb aligns with the matrix subject.

(8) *Summary: English and Mandarin trans. resultative*

a. *Canonical trans. resultative*

Matrix Verb	Matrix Subj.	Post-Verbal Obj.	English	Mandarin
Unaccus.	Add'l Subj.	Theme	✓ (4a)	✓ (7a)
Unerg.	Agent	Add'l Obj.	✓ (3b)	✓ (6b)

b. *Non-canonical trans. resultative*

Matrix Verb	Matrix Subj.	Post-Verbal Obj.	English	Mandarin
Unaccus.	Theme	Add'l Obj.	✗ (3a)	✓ (6a)
Unerg.	Add'l Subj.	Agent	✗ (4b)	✓ (7b)

3. Reappearance of unaccusative-unergative distinctions in Mandarin resultatives

Despite appearances to the contrary, Mandarin transitive resultatives do exhibit unaccusative-unergative distinctions, with respect to two diagnostics: (i) the possibility of having a corresponding BA-construction, where the post-BA NP corresponds to the post-verbal object in the transitive resultative, as schematized in (9a); and (ii) the possibility of passivization with BEI, where the subject of BEI corresponds to the post-verbal object in the transitive resultative, as schematized in (9b).

- (9) a. *BA-construction*
 NP BA NP_i V_{Cause} —_i XP_{Result}
- b. *BEI-construction*
 NP_i BEI NP V_{Cause} —_i XP_{Result}

Specifically, for transitive resultatives with an additional post-verbal object, e.g., (6ab), their corresponding BA-constructions exhibit an unaccusative-unergative distinction: (10a), where the matrix verb is unaccusative, is ill-formed, whereas (10b), where the matrix verb is unergative, is well-formed.

(10) *BA-counterparts of trans. resultative w/ add'l post-verbal obj.*

- a. ... w/ *unaccus. matrix verb*
 *Wo ba lian [jidong]-de hong-le.
 1SG BA face be.excited-DE be.red-PRF
 Int. 'I was excited, as a result my face was red.'
- b. ... w/ *unerg. matrix verb*
 Wo ba lian [ku]-de hong-le.
 1SG BA face cry-DE be.red-PRF
 'I cried, as a result my face was red.'

In addition, their corresponding BEI-constructions also exhibit an unaccusative-unergative distinction: (11a), where the matrix verb is unaccusative, is ill-formed, whereas (11b), where the matrix verb is unergative, is well-formed.

(11) *BEI-counterparts of trans. resultative w/ addt'l post-verbal obj.*

- a. ... w/ *unaccus. matrix verb* b. ... w/ *unerg. matrix verb*
 *Wo-de lian bei wo [jidong]-de hong-le. Wo-de lian bei wo [ku]-de hong-le.
 1SG's face BEI 1SG be.excited-DE be.red-PRF 1SG's face BEI 1SG cry-DE be.red-PRF
 'My face was made to be red by me, as a result of me being excited.' 'My face was made to be red by me, as a result of me crying.'

For transitive resultatives with an additional matrix subject, e.g., (7ab), their corresponding BA-constructions, (12ab), are well-formed.

(12) *BA-counterparts of trans. resultative w/ addt'l matrix subj.*

- a. ... w/ *unaccus. matrix verb* b. ... w/ *unerg. matrix verb*
 Zhe-shi ba Lisi [ji]-de bing-le. Zhe-shi ba Lisi [ku]-de bing-le.
 this-matter BA Lisi be.worried-DE be.sick-PRF this-matter BA Lisi cry-DE be.sick-PRF
 'This matter made Lisi worry, as a result (he) was sick.' 'This matter made Lisi cry, as a result (he) was sick.'

However, their corresponding BEI-constructions exhibit an unaccusative-unergative distinction: (12a), where the matrix verb is unaccusative, is well-formed, whereas (12b), where the matrix verb is unergative, is ill-formed.

(13) *BEI-counterparts of trans. resultative w/ addt'l matrix subj.*

- a. ... w/ *unaccus. matrix verb* b. ... w/ *unerg. matrix verb*
 Lisi bei zhe-shi [ji]-de bing-le. *Lisi bei zhe-shi [ku]-de bing-le.
 Lisi BEI this-matter be.worried-DE be.sick-PRF Lisi BEI this-matter cry-DE be.sick-PRF
 'Lisi was made to be worried by this matter, as a result (he) was sick.' Int. 'Lisi was made to cry by this matter, as a result (he) was sick.'

To summarize, Mandarin transitive resultatives exhibit three distinct patterns with respect to their possibility of having a corresponding BA-construction and passivization with BEI: (i) For canonical transitive resultatives (with an unaccusative or unergative matrix verb), both their BA-counterparts and their BEI-counterparts are well-formed. (ii) For non-canonical transitive resultatives with an unaccusative matrix verb and an additional post-verbal object, both their BA-counterparts and their BEI-counterparts are ill-formed. (iii) For non-canonical transitive resultatives with an unergative matrix verb and an additional matrix subject, their BA-counterparts are well-formed, but their BEI-counterparts are ill-formed. These contrasts call for a non-uniform analysis of Mandarin resultatives with unaccusative and unergative matrix verbs.

(14) *Summary: BA- and BEI-counterparts of Mandarin trans. resultative*

a. *Canonical trans. resultative*

Matrix Verb	Matrix Subj.	Post-Verbal Obj.	BA-Counterpart	BEI-Counterpart
Unaccus.	Addt'l Subj.	Theme	✓ (12a)	✓ (13a)
Unerg.	Agent	Addt'l Obj.	✓ (10b)	✓ (11b)

b. *Non-canonical trans. resultative*

Matrix Verb	Matrix Subj.	Post-Verbal Obj.	BA-Counterpart	BEI-Counterpart
Unaccus.	Theme	Addt'l Obj.	✗ (10a)	✗ (11a)
Unerg.	Addt'l Subj.	Agent	✓ (12b)	✗ (13b)

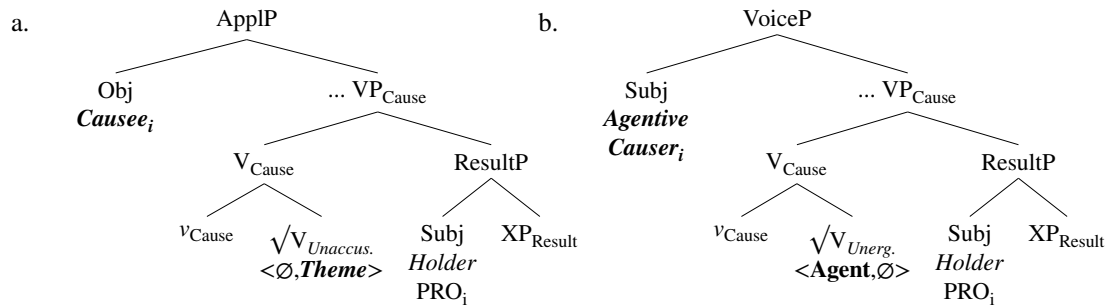
4. Proposal

4.1. A non-uniform analysis

I pursue a proposal where *UH* is upheld in (both English and Mandarin) resultatives, which I assume to consist of a matrix V_{Cause} , which is formed when a matrix verbal root is categorized by v_{Cause} , and an embedded ResultP.

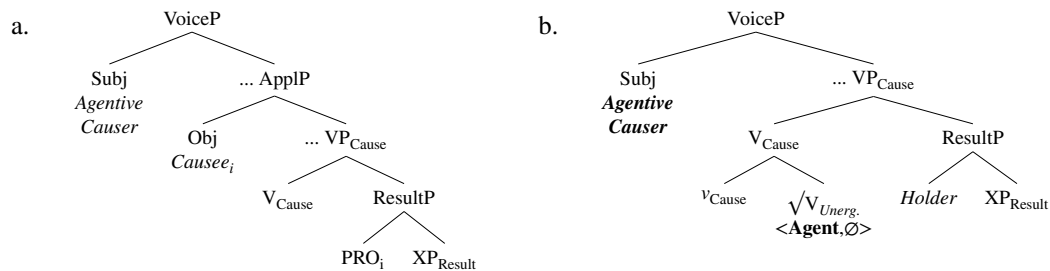
For intransitive resultatives, I propose that the unaccusative-unergative distinction lies in how the (Agent, Theme) arguments of the matrix verbal root aligns with the (*Causer*, *Causee*) arguments of V_{Cause} :² (i) for an unaccusative matrix verbal root, its Theme aligns with a Causee object of V_{Cause} , which is introduced by an Appl(icative) head, as illustrated in (15a); (ii) for an unergative matrix verbal root, its Agent aligns with an agentive Causer subject of V_{Cause} , which is introduced by Voice, as illustrated in (15b). In both structures, the subject of the embedded ResultP, which I will refer to as a *Holder*, is a PRO controlled by the sole argument of V_{Cause} . I suggest that the Mandarin-English difference lies in the principle of control that underlies *DOR*: In Mandarin, *DOR* is the result of control by the *closest argument* (cf. Huang 1989, 2006), which may be an underlying object, as in (15a), or an underlying subject, as in (15b). By contrast, in English, *DOR* is the result of control by an *underlying object*; therefore, (15b), which involves control by an underlying subject, is ruled out.

(15) *Unaccus.-Unerg. distinction in intrans. resultatives*



For transitive resultatives, I propose that both English and Mandarin allow the canonical argument structures in (16). Specifically, (i) (16a), where V_{Cause} has both an agentive Causer subject and a Causee object, which controls the PRO/Holder subject of the embedded ResultP, is derivable either from (15a) with an additional agentive Causer subject or from (15b) with an additional Causee object; (ii) in (16b), V_{Cause} has just an agentive Causer subject, which is introduced by Voice, and the post-verbal object is a Holder subject of the embedded ResultP, which is assigned case by Voice.

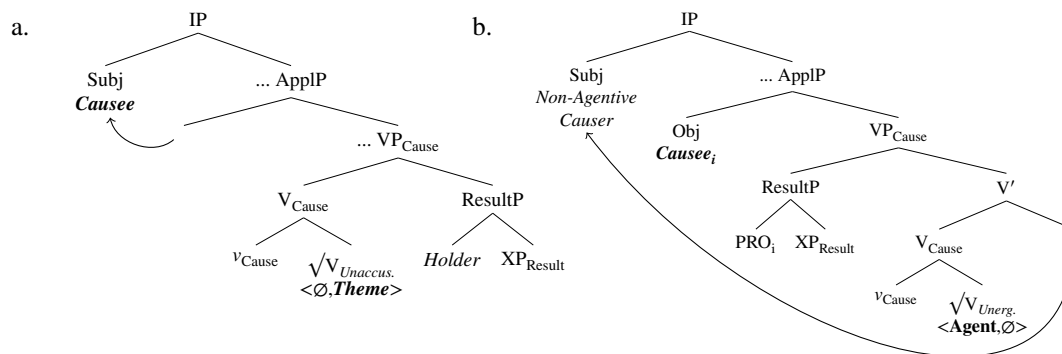
(16) *Canonical argument structures of trans. resultatives*



² In the literature, Causee is often reserved for the “logical subject of the root verb” in lexical causatives (Harley 2008) or the “embedded external argument” (Harley 2013), which is not an argument of V_{Cause} (see also Folli & Harley 2007; Legate 2014; a.o.). Here, I assume that V_{Cause} may have a Causee object, which is interpreted as being affected in the causing event, in the sense that it is caused to perform an action or undergo a change of state (as is specified by the embedded result), following Alsina (1992).

In addition, I propose that Mandarin also allows two non-canonical argument structures: (i) in (17a), V_{Cause} has just a Causee object, which is introduced by Appl and becomes the surface matrix subject, and the post-verbal object is a Holder subject of the embedded ResultP, which is assigned case by Appl; (ii) in (17b), V_{Cause} has both a Causee object, which controls the PRO/Holder subject of the embedded ResultP, and a non-agentive Causer object, which is introduced by V_{Cause} and becomes the surface matrix subject. By contrast, I suggest that in English, (17a) is ruled out because Appl cannot assign case to the Holder subject of the embedded ResultP, and (17b) is ruled out because V_{Cause} cannot introduce a non-agentive Causer object (cf. Levin & Rappaport Hovav 1995).

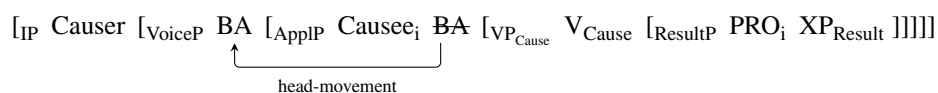
(17) *Non-canonical argument structures of trans. resultatives*



4.2. Diagnostics

Following Chen (2023; cf. Huang 1992), I take a transitive resultative's possibility of having a corresponding BA-construction as a diagnostic for affectedness/the structural position of the post-verbal object. Specifically, adopting Chen's (2023) analysis as illustrated in (18), I assume that BA spells out an Appl(icative) head that introduces the post-BA object as the Causee of the matrix V_{Cause} , which controls a PRO subject in the embedded resultative predicate (Huang 1992) and is interpreted as being affected in the causing event, in the sense that it is caused to perform an action or undergo a change of state (as is specified by the embedded result) (Alsina 1992). Therefore, a transitive resultative's lack of a well-formed BA-counterpart indicates that the post-verbal object is underlyingly the Holder subject of the embedded result.

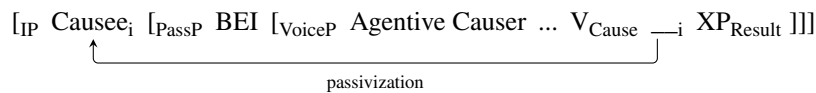
(18) *Analysis of BA-construction*



Following Chen (2023, 2025), I take a transitive resultative's possibility of passivization with BEI as a diagnostic for agentivity/the presence of an underlying matrix subject. Specifically, adopting Chen's (2023, 2025) analysis as illustrated in (19), I assume that BEI spells out a Pass(ive) head (that projects above Voice, in the sense of Bruening 2013), which requires the matrix V_{Cause} to have an underlying (agentive Causer) subject, either assigning case to it (if it is overt and introduced by Voice) or existentially closing it (if it is non-overt).³ Therefore, a transitive resultative's lack of a well-formed BEI-counterpart indicates that the matrix V_{Cause} lacks an underlying (agentive Causer) subject.

³ Chen's (2023, 2025) analysis of the BEI-construction diverges from a widely accepted alternative approach that derives the dependency involved in the BEI-construction via base-generation of the subject of BEI as an argument of BEI and $\bar{A}$ -movement of a null operator in BEI's complement, on a par with Chomsky's (1977, 1981) analysis of English tough-movement (see e.g., Feng 1995, 2012; Ting 1995, 1998; Huang 1999; Tang 2001; Huang, Li & Li 2009; Bruening & Tran 2015; a.o.). Importantly, it is only under Chen's (2023, 2025) reanalysis of the BEI-construction as a passive construction that the possibility of passivization with BEI can serve as a diagnostic for agentivity.

(19) *Analysis of BEI-construction*



The diagnostics are summarized in (20):

(20) *Summary of diagnostics*

<i>Trans. resultative has a BA-counterpart?</i>	<i>... has a passive/BEI-counterpart?</i>
✓: post-verbal obj. as Causee of matrix V _{Cause}	✓: matrix V _{Cause} has an underlying subj.
✗: post-verbal obj. as underlying subj. of embedded result	✗: matrix V _{Cause} has no underlying subj.

I propose that the canonical argument structure of transitive resultatives in (16a) and the two non-canonical argument structures of transitive resultatives in (17) derive the three distinct patterns in their possibility of having a corresponding BA-construction and passivization with BEI in Mandarin:⁴ (i) The possibility of (16a) is supported by the fact that canonical transitive resultatives (with an unaccusative or unergative matrix verb) have well-formed BA- and BEI-counterparts. In this case, when the matrix verbal root is unaccusative, its sole argument aligns with the structurally *lower* Causee object of V_{Cause}, and when the matrix verbal root is unergative, its sole argument aligns with the structurally *higher* agentive Causer subject of V_{Cause}. (ii) The possibility of (17a) is supported by the fact that non-canonical transitive resultatives with an unaccusative matrix verb and an additional post-verbal object lack well-formed BA- and BEI-counterparts. In this case, the sole argument of the unaccusative matrix verbal root still aligns with the Causee object, which is also the sole argument of V_{Cause}. (iii) The possibility of (17b) is supported by the fact that non-canonical transitive resultatives with an unergative matrix verb and an additional matrix subject have well-formed BA-counterparts but lack well-formed BEI-counterparts. In this case, the sole argument of the unergative matrix verbal root, which aligns with the Causee object of V_{Cause}, is structurally *higher* than the additional non-agentive Causer object of V_{Cause}, which becomes the surface matrix subject.

4.3. Implications

Under the proposed analysis, Mandarin's apparent lack of unaccusative–unergative distinctions in transitive resultatives, which is manifested as a flexible mapping between the (Agent, Theme) arguments of the matrix verb and the (Causer, Causee) arguments of V_{Cause}, arises because, in addition to the canonical argument structure of V_{Cause}, two non-canonical argument structures are also possible, which obscure the canonical unaccusative–unergative distinction. By contrast, in English, which exhibits apparent unaccusative–unergative distinctions in transitive resultatives and a one-to-one mapping between the (Agent, Theme) arguments of the matrix verb and the (Causer, Causee) arguments of V_{Cause}, only the canonical argument structure of V_{Cause} is possible, and the non-canonical argument structures are ruled out for independent reasons.

The proposal sheds new light on the *Uniformity of Theta-Assignment Hypothesis (UTAH)*, which posits that identical thematic relationships are represented by identical structural relationships in the argument structure (Baker 1988). In Mandarin, despite the possibility of non-canonical argument structures, an unaccusative–unergative distinction is still upheld, in the sense that (i) the sole argument of an unaccusative matrix verbal root is Causee object in intransitive resultatives and always aligns with the structurally *lower* argument of V_{Cause} in transitive resultatives, and that (ii) the sole argument of an unergative matrix verbal root is a Causer subject in intransitive resultatives and always aligns with the structurally *higher* argument of V_{Cause} in transitive resultatives.

The proposal also advances a *general theory of resultative argument structure*, where the Mandarin–English differences are derived from cross-linguistic variation in how (i) the principle of control that underlies *DOR*, (ii) the case-assigning capacity of the Applicative head that introduces the Causee, and

⁴ The possibility of the canonical argument structure of transitive resultatives in (16b) is independently supported; for evidence, see e.g., Chen (2023).

(iii) the structural position of the Causer, in particular, the possibility of a non-agentive Causer as an underlying object of V_{Cause} , are encoded in resultative constructions.

5. Against a uniform analysis of Mandarin resultatives

Contra the proposed non-uniform analysis, previous studies accept the apparent lack of unaccusative-unergative distinctions in Mandarin (intransitive and transitive) resultatives without further investigation and, as a result, abandon *UH* in favor of a *uniform* analysis of Mandarin resultatives with unaccusative and unergative matrix verbs (see e.g., Huang 1988, 2006; Sybesma 1992, 1999; a.o.).

In a representative proposal by Huang (1988, 2006), intransitive resultatives are analyzed as non-causative/inchoative constructions, as illustrated in (21), in which (i) a Become head selects a stative predicate and introduces a change-of-state event, (ii) the Become head may be modified by either an unaccusative or unergative matrix verb, and (iii) the sole argument of the unaccusative or unergative matrix verb is identified with the subject of the stative predicate; and transitive resultatives are analyzed as causative constructions, as illustrated in (22), involving (i) both a Cause head, which introduces a causing event and may be modified by an unergative matrix verb, as in (22a), or be ‘pure’/unmodified, as in (22b) and (22c), and a Become head, which introduces the caused change-of-state event and may be modified by either an unaccusative or unergative matrix verb, and (ii) an additional post-verbal object identified with the subject of the stative predicate or an additional matrix subject introduced by the Cause head.⁵

(21) Huang’s (2006) analysis of intrans. resultative

- a. ... w/ *unaccus. matrix verb*
[Become_{<unaccus.>} [*x* <State>]]
e.g., (5a) = Lisi became sick *from being worried*.
- b. ... w/ *unerg. matrix verb*
[Become_{<unerg.>} [*x* <State>]]
e.g., (5b) = Lisi became sick *from crying*.

(22) Huang’s (1988, 2006) analysis of trans. resultative

- a. ... w/ *unerg. matrix verb & addt’l post-verbal obj.*
[*x* Cause_{<unerg.>} [Become [*y* <State>]]]
e.g., (6b) = I caused *by crying my face* to become red.
- b. ... w/ *unaccus. matrix verb & addt’l matrix subj.*
[*x* Cause [Become_{<unaccus.>} [*y* <State>]]]
e.g., (7a) = This matter caused me to become sick *from being worried*.
- c. ... w/ *unerg. matrix verb & addt’l matrix subj.*
[*x* Cause [Become_{<unerg.>} [*y* <State>]]]
e.g., (7b) = This matter caused me to become sick *from crying*.

⁵ Huang (1988, 2006) does not provide an analysis for transitive resultatives with an unaccusative matrix verb and an additional post-verbal object (e.g., 6a). But it is conceivable that his analysis for transitive resultatives with an unergative matrix verb and an additional post-verbal object, as illustrated in (22a), can be extended to the unaccusative case, with the *Cause* head being modified by an unaccusative matrix verb, as illustrated in (i).

- (i) *Trans. resultative w/ unaccus. matrix verb & addt’l post-verbal obj.*
[*x* Cause_{<unaccus.>} [Become [*y* <State>]]]
e.g., (6a) = I caused *by being worried my face* to become red.

According to Huang (1988, 2006), The Mandarin-English difference is that in Mandarin resultatives, either an unaccusative or unergative verb can modify the Become head (and the Cause head, if taken into consideration the analysis in (i)), whereas in English resultatives, only an unergative verb can modify the Become head and the Cause head.

Such a uniform analysis faces an immediate challenge: As shown in section 3, transitive resultatives exhibit three distinct patterns with respect to their possibility of having a corresponding BA-construction and passivization with BEI, which, under a uniform analysis, remain unexplained.

In addition, contra Huang's (1988, 2006) analysis in (21), I argue that intransitive resultatives are not non-causative/inchoative constructions: As seen in (23), when the matrix verb and the embedded result have independent temporal modification, *jintian* 'today', which modifies the embedded result, *cannot* occur sentence-initially (in the matrix clause, where *zuotian* 'yesterday' modifies the matrix verb). This is expected under a causative analysis (which involves both a matrix Cause head and an embedded result) and unexpected under a non-causative/inchoative analysis where the Become head is in the matrix clause.

- (23) a. *Temporal modification in intrans. resultative*
 {*Jintian,} Lisi zuotian [ji] [ku]-de {jintian} bing-le.
 today Lisi yesterday be.worried/cry-DE today be.sick-PRF
 Causative paraphrase: '{*Today,} Lisi was worried/cried yesterday, as a result (he) got sick {today}.'
 Inchoative paraphrase: '{Today,} Lisi became sick {today} from being worried/crying yesterday.'
- b. *Cf. because-clause adjunction in unaccus./unerg. construction*
 {Jintian,} Lisi [yinwei (ta) zuotian hen zhaoji] [ku]-le {jintian} bing-le.
 today Lisi because 3SG yesterday DEG be.worried/cry-PRF today be.sick-PRF
 '{Today,} Lisi, because (he) was worried/cried yesterday, got sick {today}.'

Furthermore, contra Huang's (1988, 2006) analysis in (22b) and (22c), I argue that transitive resultatives with an additional matrix subject are not 'pure' causatives where the matrix verb modifies a Become head instead of a Cause head: As seen in (24), a selectional restriction is imposed on the additional matrix subject by the matrix verb, such that it must be identified with a source of emotion/cognition (e.g., what Lisi was excited/crying about). This contrasts with the *shi*-construction in (25), which is arguably a 'pure' causative, where the causative head, spelled out by *shi* 'make', is unmodified.

- (24) *Non-agentive Causer as source of emotion/cognition in non-canonical trans. resultative*

- a. *Xingfenji [jidong]-de Lisi bing-le.
 stimulant be.excited-DE Lisi be.sick-PRF
 INT: 'The stimulant made Lisi excited, as a result (he) was sick.'
- b. *Cuileidan [ku]-de Lisi bing-le.
 tear gas cry-DE Lisi be.sick-PRF
 INT: 'The tear gas made Lisi cry, as a result (he) was sick.'

- (25) *Cf. Shi-construction as 'pure' causative*

- a. Xingfenji shi Lisi [jidong]-de bing-le.
 stimulant make Lisi be.excited-DE be.sick-PRF
 'The stimulant made Lisi excited, as a result (he) was sick.'
- b. Cuileidan shi Lisi [ku]-de bing-le.
 tear gas make Lisi cry-DE be.sick-PRF
 'The tear gas made Lisi cry, as a result (he) was sick.'

6. Conclusion

In conclusion, I have shown that, despite appearances to the contrary, Mandarin transitive resultatives do exhibit unaccusative-unergative distinctions with respect to their possibility of having a corresponding

BA-construction and passivization with BEI. I have proposed a non-uniform analysis of resultatives with unaccusative and unergative matrix verbs, in accordance with *UH*, which not only sheds new light on *UTAH*, but also advances a general theory of resultative argument structure.

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